

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (IT) Examination

May 2014

IT207 COMPUTER NETWORKING-I

Date: 05.05.2014, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain duties of Transport Layer in OSI Model. [05]
(b) Draw TCP/IP Protocol suit Model and explain network and physical layers. [05]
- Q - 2 (a) What is DTE? Explain Data Communication Equipment. [06]
(b) Choose the best matching between group1 and group2 [03]

Group - 1	Group - 2
P. Data link layer	1. Ensures reliable transport of data over a physical point-to-point link
Q. Network layer	2. Encodes / decodes data for physical transmission
R. Transport layer	3. Allows end-to-end communication between two processes
	4. Routes data from one network node to the next

- (c) Describe random access protocol CSMA/CA in detail [06]

OR

- Q - 2 (a) Compare Circuit Switch Network Vs Datagram Packet Network. [05]
(b) List out and Explain any two networking devices which can be configured on Data Link Layer. [05]
(c) List out noisy and noiseless protocols. Explain Selective Repeat ARQ Protocol. [05]

- Q - 3 (a) Find out whether there is an error or not at receiver side for the data word = 101100 using CRC error detection technique. (Given: divisor = 1101 and redundant bits= 001. [07]
(b) What is the size of sender window and receiver window in all noisy protocol? [03]

OR

- Q - 3 (a) Find the redundant bits for the following bit pattern with LRC [06]
11100111 11011101 0011001 10101001
(b) What is Bit Stuffing? Explain baud rate. [04]

SECTION – II

- Q - 4 (a) Write a short note on cable modem. [06]
(b) Draw the pattern for Manchester encoding for 101001110101 bits. [04]
- Q - 5 (a) Explain Sliding Window mechanism. [06]
(b) How link state routing differs from distance vector routing? [05]
(c) Discuss Transmission Impairments. [05]
- OR
- Q - 5 (a) Compare Guided Vs. Unguided Transmission Medium. [06]
(b) Explain Polar encoding techniques with example. [05]
(c) What is Modulation? Discuss Frequency Modulation. [05]
- Q - 6 (a) Explain Time Division Multiplexing. [05]
(b) What are the responsibilities of the Data Link Layer? [04]
- OR
- Q - 6 (a) Discuss Null Modem Communication. [05]
(b) Compare Periodic Vs. Aperiodic Signals. [04]